Ch. 7 CW #1

Principles of Microeconomics

October 14, 2025

1. Consider a market where supply is given by $Q_S = 10P - 40$ and demand is given by $Q_D = 200 - 10P$.

   (a) What is the market equilibrium price and quantity?

   (b) What is consumer surplus at the market equilibrium?

   (c) What is producer surplus at the market equilibrium?

   (d) If the government implements a price floor of \$15, what quantity of the good gets traded?

   (e) In the world of the price floor, what are consumer surplus and producer surplus?

   (f) In the world of the price floor, what is deadweight loss?

   (g) If the government instead implements a price ceiling of \$9, what quantity of the good gets traded?

   (h) In the world of the price ceiling, what are consumer surplus and producer surplus?

   (i) In the world of the price ceiling, what is deadweight loss?

   (j) If the government instead implements a tax of \$4 per good sold on suppliers, what price do consumers end up paying?

   (k) In the world of the tax, what price do suppliers end up receiving

   (l) In the world of the tax, what quantity of the good gets traded?

   (m) In the world of the tax, what are consumer surplus, producer surplus, and tax revenue?

   (n) In the world of the tax, what is deadweight loss?